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Academic Year 2026 - 27

S5 PROJECT WORK I

First Review

DRUG INVENTORY AND SUPPLY CHAIN TRACKING SYSTEM

BIP PROJECT ID
2026MIN549

CATEGORY
INTERNAL

BATCH MEMBERS

Dayanand P C (7376241ME109)

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GUIDE

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LITERATURE SURVEY

Sl.No.	References	Works carried out	Information gathered relevant to your project
1	X. Chen, G. Lu, H. Zhang, and J. Wan, "Knowledge graph-enhanced deep learning for pharmaceutical demand forecasting," Sci. Rep., 2026, doi: 10.1038/s41598-026-35113-4.	The authors built a model called KG-GCN-LSTM. KG stands for Knowledge Graph. GCN stands for Graph Convolutional Network. The Knowledge Graph part links related drugs together. The Graph Convolutional Network part reads those links.	Provides base layer of technology. The GCN as well as KG together give the required foundation to understand the cross-product and cross category tracking.
2	R. Pall, Y. Gauthier, S. Auer, and W. Mowaswes, "Predicting drug shortages using pharmacy data and machine learning," Health Care Management Science, vol. 26, 2023, doi: 10.1007/s10729-022-09627-y.	The authors used sales data from 22 Canadian pharmacies. They built machine learning models to guess if a drug shortage is coming. They sorted shortages into four levels: none, low, medium, high.	The model correctly guessed the shortage level 69 percent of the time, one month ahead. It did this without any stock data from drug makers or suppliers.
3	K. Yahya, M. Safaei, and S. Al Dawsari, "Hybrid machine learning forecasting for resilient and sustainable pharmaceutical supply chains under regulatory and seasonal disruption," Frontiers in Artificial Intelligence, vol. 9, article 1803863, 2026, doi: 10.3389/frai.2026.1803863.	Support Vector Regression handled the steady, normal part of drug demand. Deep Neural Network handled the sudden ups and downs. They tested this on real demand data .They also used SHAP (SHapley Additive exPlanations) to explain why the model made each prediction.	The combined model had lower error than every other model tested. The paper states directly that this kind of model can plug into an existing ERP dashboard for real-time decisions.
4	S. Mabizela, H. N. Nakambale, and V. Bangalee, "Predictors of medicine redistribution at public healthcare facilities in King Cetshwayo District, KwaZulu-Natal, South Africa," BMC Health Services Research, vol. 23, article 1108, 2023, doi: 10.1186/s12913-023-10096-4.	The authors looked at pharmacy stock records from 8 healthcare facilities in South Africa. They checked 392 medicines in total. They used statistics to find the link between running out of stock, having too much stock, and stock going to waste from expiry.	Informed real-time batch/expiry data-capture statistics so dashboard calculations can be based on it.
5	R. Uthayakumar and S. Priyan, "Pharmaceutical supply chain and inventory management strategies: Optimization for a pharmaceutical company and a hospital," Operations Research for Health Care, vol. 2, no. 3, pp. 52 to 64, 2013.	The authors built math models for stock control in a two-step drug supply chain. One step is the drug maker. The other step is the hospital. The models account for the cost of running out of stock, the cost of holding stock, and patient safety limits.	This foundational paper provides the important insight on stakeholder identification, whether target is one or multiple nodes in a chain.

LITERATURE SURVEY

Sl.No.	References	Works carried out	Information gathered relevant to your project
6	R. K. Khokrale, "Using cloud-based ERP solutions for real-time supply chain visibility and decision-making," International Journal of Research Publications in Engineering, Technology and Management, vol. 7, no. 2, pp. 9811 to 9830, 2024, doi: 10.15662/IJRPETM.2024.0702003.	The goal was to understand how cloud-based ERP systems affect supply chain visibility. Visibility as how clearly a company can see what is happening across its supply chain in real time.	The paper states that cloud ERP lists responsiveness, data quality, and operational agility as the main benefits.
7	T. K. Agrawal, R. Kalaiarasan, J. Olhager, and M. Wiktorsson, "Supply chain visibility: A Delphi study on managerial perspectives and priorities," International Journal of Production Research, vol. 62, no. 8, pp. 2927 to 2942, 2024, doi: 10.1080/00207543.2022.2098873.	The authors ran a Delphi study. They asked 26 supply chain experts from global companies about what matters most for supply chain visibility.	The paper names cloud-based systems as one of the key technologies that experts rely on to get this visibility, prioritising visibility over forecasting.
8	F. Balaha, H. Albinali, H. Alrabiah, M. Ali, and Z. Bahroun, "An analytical review of data integration for decision support in smart manufacturing," Decision Analytics Journal, vol. 17, article 100647, 2025.	The authors reviewed how data flows from factory sensors up to ERP systems. This is called the IIoT-to-ERP path. IIoT stands for Industrial Internet of Things.	The paper finds that most systems assume real-time data will be available at the decision-making level, but this is rarely tested in real use. It names network delay limits, message queue design, and tagging of data meaning as the three most important engineering factors for this to work.
9	M. Sharma et al., "AI integration in ERP evaluation across trends and architectures," Journal of Information System Engineering and Management, 2025.	The authors reviewed how Artificial Intelligence is built into cloud ERP systems, including SAP S/4HANA, Oracle Fusion, and Microsoft Dynamics 365.	The paper finds there is no standard method yet to measure how well Artificial Intelligence automation works inside an ERP system. This means results reported by different companies cannot be fairly compared to each other.
10	K. J. Harianto, Z. J. H. Tarigan, H. Siagian, S. R. Basana, and F. Jie, "The effect of digital ERP implementation, supply chain integration and supply chain flexibility on business performance," *International Journal of Data and Network Science*, vol. 8, no. 4, pp. 2399 to 2414, 2024.	The authors built a statistical model to test how ERP setup, supply chain integration, and supply chain flexibility connect to business results.	The paper finds that supply chain integration is the missing link between installing ERP and actually seeing better business performance.

AIM & OBJECTIVES OF THE PROJECT (Problem Statement)

- **Aim:** To design and develop a cloud-based predictive analytics dashboard for drug inventory and supply chain tracking, capable of forecasting inventory imbalance, generating a composite safety score per SKU/batch, and triggering real-time alerts, thereby shifting drug stock management from reactive to predictive.
- **Objectives**
  - Design a cloud-native data architecture on cloud-hosted MySQL. Engineer it for real-time synchronization and horizontal scalability, to serve as the unified data layer for downstream analytics.
  - Develop an imbalance-detection model, using techniques such as reorder-point thresholds, moving averages, and expiry-window rules, to flag stockout risk and expiry-driven wastage.
  - Engineer a composite safety-scoring mechanism that aggregates current stock level, days-to-expiry, and demand variability into a single risk index per SKU/batch.
  - Validate the end-to-end system against synthetic transaction data. Measure stockout reduction and expiry-wastage reduction to establish proof of concept ahead of a production-scale, ML-extendable deployment path.

SCOPE OF THE PROJECT

- Covers rule-based drug inventory tracking at SKU/batch level for a single pharmacy or hospital stock point, using threshold, moving-average, and expiry-window logic (no ML) to flag stockout and wastage risk in real time.
- Delivered as a cloud-hosted (MySQL) web dashboard, validated on synthetic transaction data in place of access-restricted real pharmacy records.
- Multi-site/supplier integration, procurement automation, and regulatory compliance (e.g., drug-authority tracking, HIPAA-equivalent audits) are out of scope for this phase.
- Targets a single real-world use case, a hospital or retail pharmacy drug store, where stockouts and expiry wastage are the primary daily risks.
- Structured to extend into ML-based forecasting and multi-node supply chain visibility in future iterations.

NEED FOR THE CURRENT STUDY

- Manual and spreadsheet-based drug inventory tracking delays visibility into stock levels. This delay is critical given medicines have a fixed shelf life, unlike typical retail stock.[7]
- Stockouts, overstocking, and expiry-related wastage are widespread and interlinked problems in real pharmacy settings, not isolated or rare failures.[2][5]
- Existing cloud ERP and inventory-visibility solutions are largely generic or manufacturing-focused. Drug-specific inventory dynamics, expiry-driven risk and patient-safety stakes, remain underserved by a lightweight, deployable system.[3][6]
- Current options sit at two extremes: full ERP suites, too heavy for single-pharmacy scale, or manual registers, too slow for real-time risk detection. There is a gap for a mid-tier, rule-based, cloud-hosted system sized for a single pharmacy or hospital stock point.
- A real-time safety-score and alert system can catch stockout and expiry risk before it becomes a patient-care or financial issue, directly addressing the pattern above.[2]

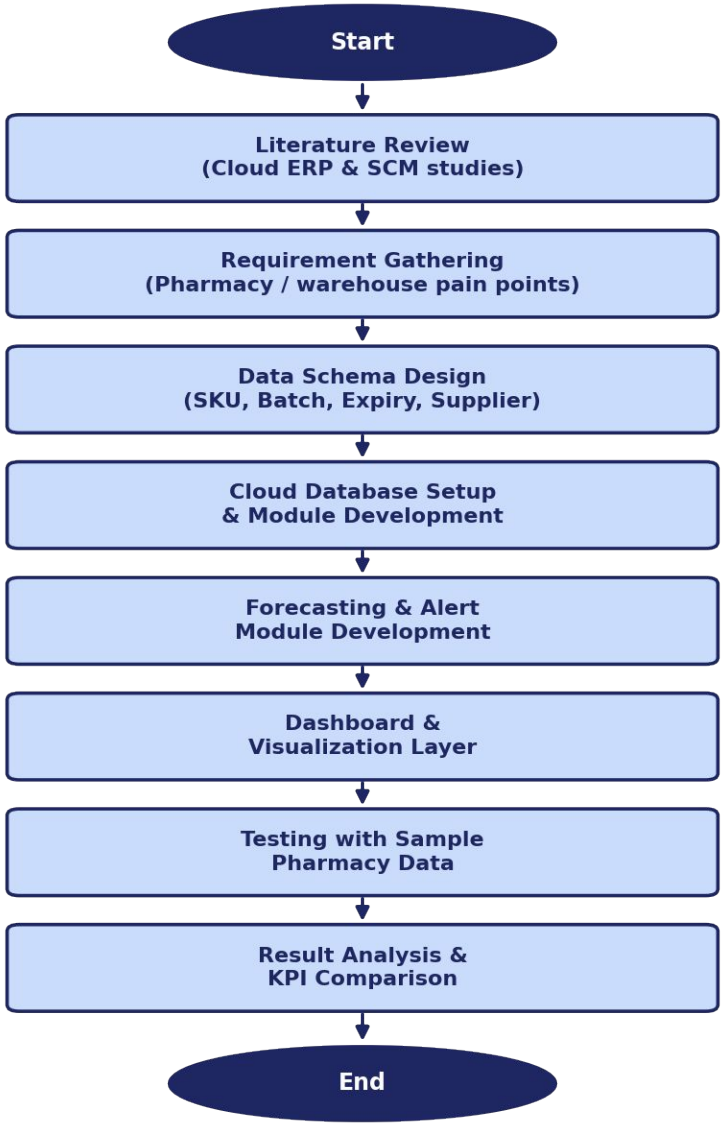
FEASIBILITY ANALYSIS

- **Technical Feasibility**
  - Built on low-cost cloud infrastructure: cloud-hosted MySQL for real-time data sync and a lightweight responsive web dashboard for stock visibility [6]
  - Stack choice prioritizes horizontal scalability and minimal DevOps overhead - suited to academic-scale deployment with a clear upgrade path to production
- **Economic Feasibility**
  - Free-tier cloud services (AWS Student plan) keeps infrastructure cost within a student-project budget, with near-zero upfront capital expenditure
  - Literature on cloud ERP adoption reports a significant improvement of ROI once usage scales, driven by reduced stockouts, lower manual audit effort, and reduced wastage from expired drugs leading to the project scope [3]
- **Operational Feasibility**
  - A minimal UI lowers the training burden for pharmacy staff with variable digital literacy
  - Prior research shows that implementation outcomes are driven primarily by organizational readiness and user training rather than system functionality.

FEASIBILITY ANALYSIS

- **Schedule Feasibility**
  - Modular, phased development plan (schema design → core CRUD → barcode integration → dashboard → testing) fits comfortably within the 8-week mini-project timeline, with each module independently testable before integration [10]
- **Risk Feasibility**
  - **Data-governance risk:** Mismatched SKUs, batch numbers, or expiry formats are a common failure point in inventory-system integrations. Mitigated by locking a standardized schema — batch ID, SKU, expiry date, quantity, drug name — before development begins
  - **Data-access risk:** Real pharmacy inventory data is access-restricted, so the prototype uses synthetic transaction data (batch cycles, expiry timelines, dispensing volumes) to validate core functionality
- **Legal/Compliance Feasibility**
  - No real patient data is collected, stored, or processed at any stage — all test data used for validation is calculated, ensuring the prototype carries no PHI/PII exposure risk and stays clear of healthcare data-privacy regulations (e.g., HIPAA-equivalent concerns) during development

PROPOSED METHODOLOGY (Flow Chart)



CHOICE OF COMPONENTS / MODULES / METHODS/TECHNIQUES EQUIPMENT USED FOR PROJECT DEVELOPMENT

- **Data & Storage Layer**
  - MySQL database hosted on a single AWS EC2 instance as the single source of truth for stock data, chosen to keep infrastructure consolidated and cost-efficient for the prototype phase.
  - Schema fields: Drug/SKU ID, Batch Number, Expiry Date, Quantity, Supplier ID, Location [9].
  - This schema-first approach is grounded in literature showing that getting data governance right before implementation matters more than the underlying platform for long-term performance [8].

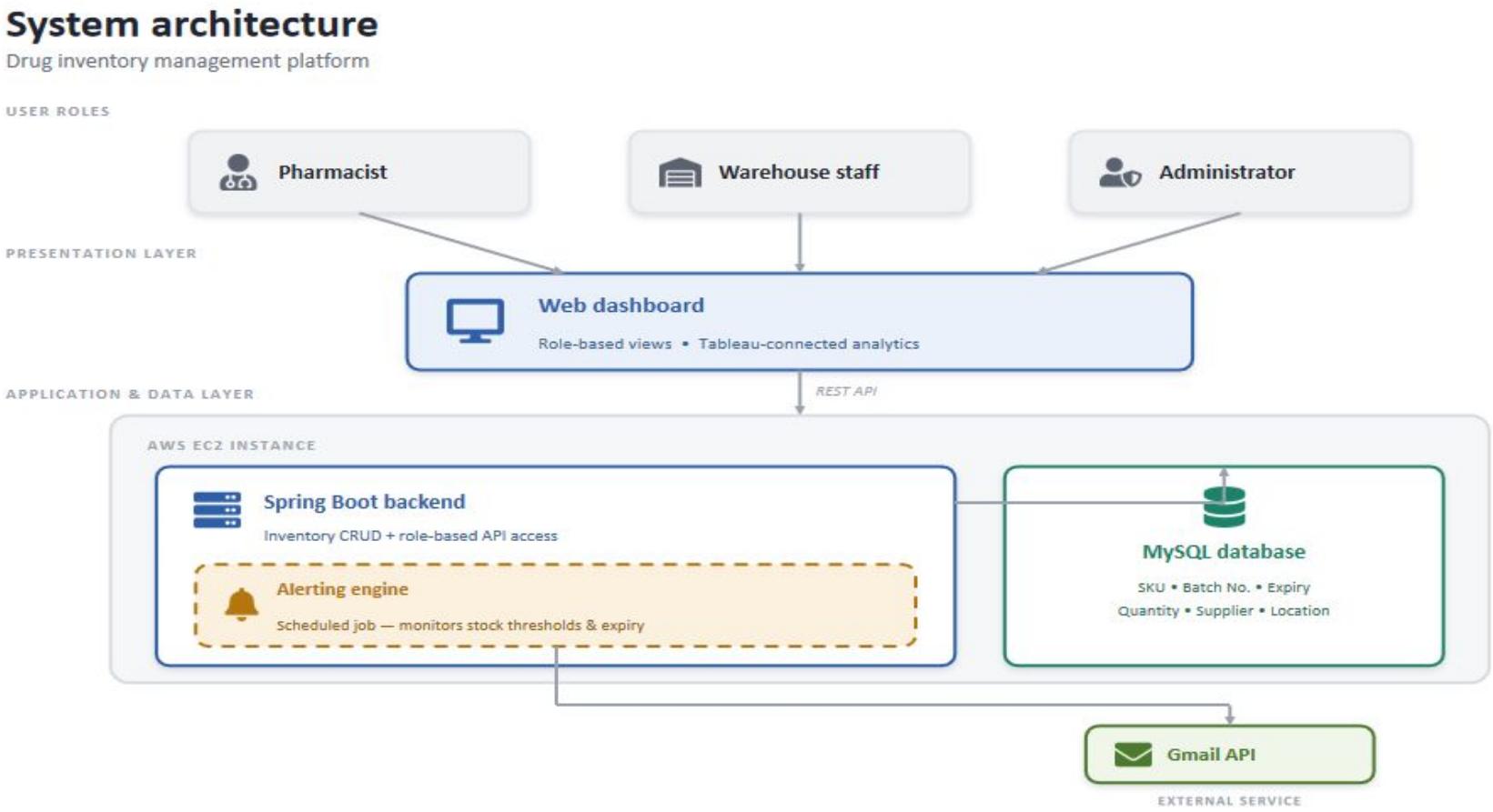
- **Application & Analytics layer**
  - Backend service built with Spring Boot, exposing inventory CRUD and alerting APIs, co-hosted on the same AWS EC2 instance as the MySQL database.
  - ML-based demand-forecasting module trained on historical dispensing patterns (batch consumption, seasonal trends, drug-specific usage rates) to predict future stock requirements and automatically trigger reorder alerts before stockouts occur. [1]
  - Visualization layer using Tableau for stock, expiry, and consumption trend analytics.

CHOICE OF COMPONENTS / MODULES / METHODS/TECHNIQUES EQUIPMENT USED FOR PROJECT DEVELOPMENT

- **Interface & Alert Layer**
  - Automated alerting engine that continuously monitors stock thresholds and expiry timelines, triggering email/SMS notifications via a third-party gateway (e.g., Gmail API) when low-stock or near-expiry conditions are detected
  - Role-based access control (RBAC) enforced at the API layer, with distinct permission scopes for pharmacist, warehouse staff, and administrator roles — ensuring users only access data and actions relevant to their function

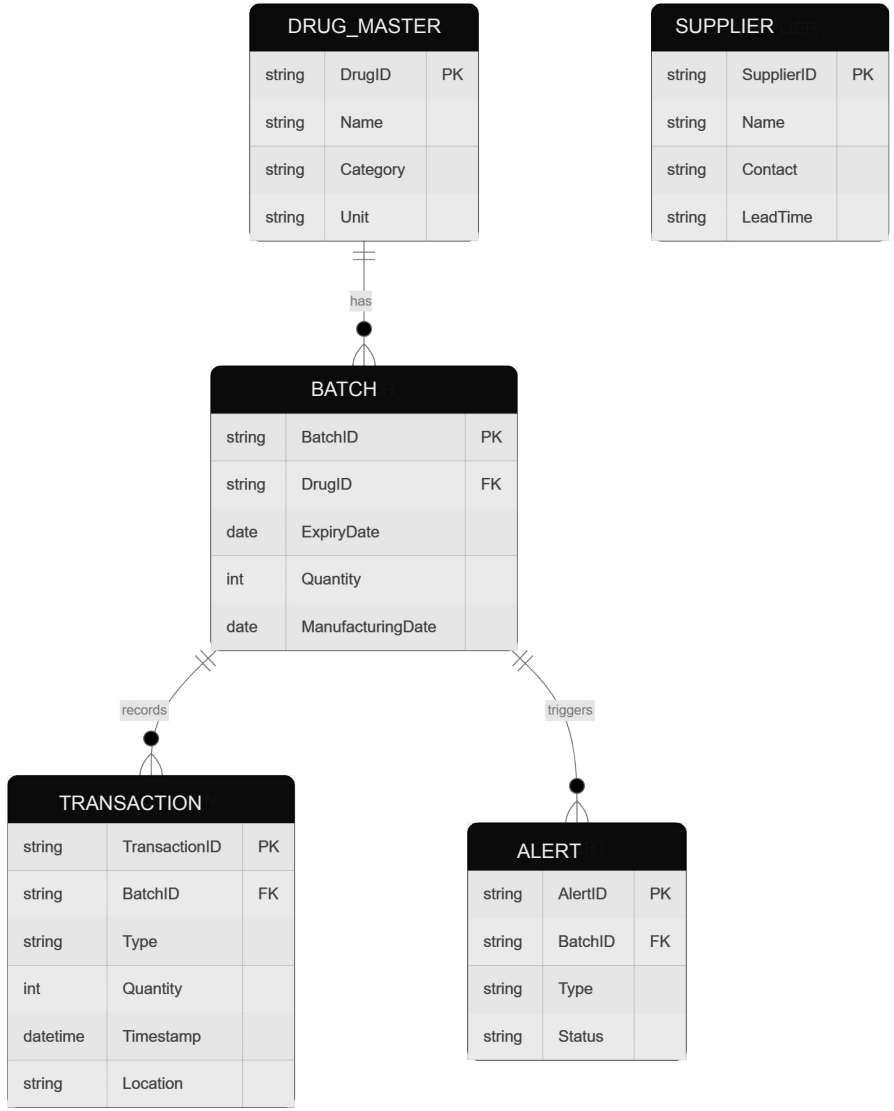
DESIGN(S) (HARDWARE / SOFTWARE ARCHITECTURE)

- **System Architecture (Layered)**



DESIGN(S) (HARDWARE / SOFTWARE ARCHITECTURE)

- **Database Schema (Entity-Relationship Design)**
  - Drug Master: Drug ID, Name, Category, Unit
  - Batch: Batch ID, Drug ID (FK), Expiry Date, Quantity, Manufacturing Date
  - Supplier: Supplier ID, Name, Contact, Lead Time
  - Transaction: Transaction ID, Batch ID (FK), Type (Inbound/Outbound), Quantity, Timestamp, Location
  - Alert: Alert ID, Batch ID (FK), Type (Low-Stock/Near-Expiry), Status
- **Dashboard / UI Design**
  - Stock Overview panel with colour-coded status
  - Expiry Alerts panel
  - Reorder Suggestions panel
  - Searchable history of all inbound/outbound stock movements



INDIVIDUAL CONTRIBUTIONS TO THE WORK
DAYANAND P C (7376241ME109)

- Conducted the literature survey across 15 papers, covering pharmaceutical demand forecasting, inventory imbalance detection, cloud ERP visibility, and supply chain analytics, and mapped findings to the project's need and objectives.
- Compiled the Need for the Current Study and Scope of the Project sections, grounding them in documented evidence (stockout, overstocking, and expiry-wastage patterns from reviewed literature).
- Structured the Aim and Objectives, and defined the composite safety-score concept (stock level, days-to-expiry, demand variability) as the project's core theoretical model.
- Prepared the project's overall documentation and review presentation, including the proposed methodology (flowchart and Gantt chart) and reference compilation in IEEE format.
- Planned the phased development approach (schema design, core CRUD, dashboard, testing) and sequenced tasks across the 8-week timeline.

INDIVIDUAL CONTRIBUTIONS TO THE WORK
HARISH S S (7376241EC169)

- Studied and evaluated technical, economic, operational, schedule, risk, and legal/compliance feasibility, including comparison of stack options (cloud-hosted MySQL, AWS free-tier) with cost and effort justification.
- Drafted the entity-relationship schema design (Drug Master, Batch, Supplier, Transaction, Alert) and proposed field-level standards (SKU, batch ID, expiry date, quantity) to prevent data-governance mismatches.
- Researched cloud database architecture options for real-time stock synchronization and outlined the planned CRUD (Create, Read, Update, Delete) operation structure.
- Studied rule-based imbalance-detection approaches (reorder-point thresholds, moving-average consumption checks, expiry-window rules) and outlined the planned alert-triggering logic.
- Identified synthetic data requirements (batch cycles, expiry timelines, dispensing volumes) for future validation of the backend-to-dashboard pipeline.

PENDING WORKS AND PLAN FOR COMPLETION

- Implement the finalized entity-relationship schema on cloud-hosted MySQL and validate real-time data synchronization.
- Develop core CRUD operations and connect them to the rule-based imbalance-detection logic (reorder point, moving average, expiry window).
- Build the composite safety-score calculation and wire it to the alert-triggering engine.
- Develop the web dashboard interface and integrate it with the backend pipeline for live stock and alert visibility.
- Validate the mid-prototype against synthetic transaction data and expand the literature base to cover any gaps identified during implementation.

PLAN FOR PUBLICATIONS

- **Journal / Conference Identified for Submission**
  - Target: UG-level Supply chain related conference focusing on digitalization and Cloud implementation.
- **Tentative Dates for Submission / Presentation / Acceptance**
  - Submission planned after Initial prototype design and literature validation.

Status on Partial Completion and Submission of Project Report

List of Documents to be Submitted

SL.No	List of Documents	Status (Provide the drive link of prepared document)
1	Cover Page & Title Page (Both are in same format)	Completed
2	Bonafide Certificate	Completed
3	Declaration	Completed
4	Acknowledgement	Completed
5	Chapter I – Introduction	Completed
6	Chapter 2 – Literature Survey	Completed

REFERENCES
(Journal Papers/ Books/ Website References)

- **As per IEEE Format**

- [1] X. Chen, G. Lu, H. Zhang, and J. Wan, “Knowledge graph-enhanced deep learning for pharmaceutical demand forecasting,” Scientific Reports, vol. 16, Art. no. 4776, 2026, doi: 10.1038/s41598-026-35113-4.
- [2] R. Pall, Y. Gauthier, S. Auer, and W. Mowaswes, “Predicting drug shortages using pharmacy data and machine learning,” Health Care Management Science, vol. 26, no. 3, pp. 395–411, 2023, doi: 10.1007/s10729-022-09627-y.
- [3] K. Yahya, M. Safaei, and S. Al Dawsari, “Hybrid machine learning forecasting for resilient and sustainable pharmaceutical supply chains under regulatory and seasonal disruption,” Frontiers in Artificial Intelligence, vol. 9, Art. no. 1803863, 2026, doi: 10.3389/frai.2026.1803863.
- [4] S. Mabizela, H. N. Nakambale, and V. Bangalee, “Predictors of medicine redistribution at public healthcare facilities in King Cetshwayo District, KwaZulu-Natal, South Africa,” BMC Health Services Research, vol. 23, Art. no. 1108, 2023, doi: 10.1186/s12913-023-10096-4.
- [5] R. Uthayakumar and S. Priyan, “Pharmaceutical supply chain and inventory management strategies: Optimization for a pharmaceutical company and a hospital,” Operations Research for Health Care, vol. 2, no. 3, pp. 52–64, 2013, doi: 10.1016/j.orhc.2013.08.001.

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- [6] R. K. Khokrale, “Using cloud-based ERP solutions for real-time supply chain visibility and decision-making,” International Journal of Research Publications in Engineering, Technology and Management, vol. 7, no. 2, pp. 9811–9830, 2024, doi: 10.15662/IJRPETM.2024.0702003.
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- [8] F. Balaha, H. Albinali, H. Alrabiah, M. Ali, and Z. Bahroun, “An analytical review of data integration for decision support in smart manufacturing,” Decision Analytics Journal, vol. 17, Art. no. 100647, 2025, doi: 10.1016/j.dajour.2025.100647.
- [9] M. Sharma, “AI integration in ERP evaluation across trends and architectures,” Journal of Information Systems Engineering and Management, vol. 10, no. 61s, 2025.
- [10] K. J. Harianto, Z. J. H. Tarigan, H. Siagian, S. R. Basana, and F. Jie, “The effect of digital ERP implementation, supply chain integration and supply chain flexibility on business performance,” International Journal of Data and Network Science, vol. 8, no. 4, pp. 2399–2414, 2024, doi: 10.5267/j.ijdns.2024.5.017.